

MLPerf(TM) Storage v3.0 - System Description

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System name: aws_r5n24xlarge_i3en12xlarge_4c_10s_64p
Workload: Checkpointing - llama3-70b, 64 MPI processes (CLOSED, 16/client)
Storage system: Alluxio Enterprise AI 3.9 (RC18) over AWS S3, via Alluxio-FUSE (POSIX)
Result: Save 34.49 GiB/s / Load 34.08 GiB/s (10 checkpoints; two-invocation split)
Cloud: AWS (us-west-1), single availability zone

1. Hardware configuration & node counts

Clients (host nodes): 4 x r5n.24xlarge

- 4 FUSE checkpoint clients, 16 MPI processes each (64 total)
- CPU: 96 vCPU (Intel Xeon Platinum 8259CL) Memory: 768 GiB
- Storage: EBS gp3 (root) Network: 100 Gbps ENA

Storage nodes: 10 x i3en.12xlarge

- 10 Alluxio storage workers (NVMe page store) + distributed FoundationDB
- CPU: 48 vCPU (Intel Xeon Platinum 8259CL) Memory: 384 GiB
- Storage: 4x 7.5 TB NVMe SSD -> RAID0 (~28 TB) page store Network: 50 Gbps ENA
- FoundationDB: distributed fdbserver across worker NVMe (configure new double ssd, coordinators auto)

2. System & network topology

Benchmark access is POSIX/file via Alluxio-FUSE; AWS S3 is the under-file-system (UFS). All nodes in one AWS us-west-1 availability zone over 50/100 Gbps ENA.

```
4x r5n.24xlarge  --Alluxio FUSE (POSIX, write)--> 10x i3en.12xlarge workers
(MPI, 16 procs ea)    write-back cache          NVMe page store + distributed FDB
                                     |
                                     AWS S3 (UFS, TRANSIENT ~0)
```

3. Software stack & versions

- Alluxio Enterprise AI 3.9 (RC18) (distributed cache + write-back)
- Alluxio-FUSE (POSIX client)
- FoundationDB 7.3.63 (write-cache metadata)
- etcd 3.5.x (mount table + worker membership)
- Ubuntu 24.04 LTS
- OpenJDK 17
- mlpstorage 3.0.x (uv-managed) + DLIO 3.0.x
- Backing store: AWS S3 (UFS)

4. Non-default configuration options

- Client write cache: alluxio.write.cache.enabled=true + FoundationDB (FdbBasedFileSystem)
- Checkpoint path routed to policyMode: TRANSIENT (write-back cache; S3 persist ~0 during run)
- Multipart write buffer: part.size=64MiB, buffer.number=512, threads=512, max.buffers.per.stream=32
- alluxio.dora.client.ufs.fallback.enabled=true; list.status.from.ufs.enabled=false
- worker.network.netty.file.transfer=TRANSFER; frame.decoder.buffer.cumulator=COMPOSITE
- FUSE JVM: -Xmx96g -XX:MaxDirectMemorySize=48g
- Two-invocation split: write (10/0) -> drop_caches -> read (0/10); inter-phase gap <=30s (rules 4.7.1)

5. Notes

Simulated accelerators (no GPU). Machine-readable system parameters are in the companion systemname.yaml. CLOSED workload parameters per mlpstorage; storage-system configuration is unconstrained in CLOSED. Dataset backing store: AWS S3.